Ethical Auditing & Documentation Debt Mitigation

1. Introduction

This project aims to design a speech processing pipeline that simultaneously addresses two critical concerns in modern AI systems: **privacy preservation** and **fairness**. Specifically, the system modifies speech signals to obfuscate sensitive biometric attributes such as gender and age, while ensuring that the linguistic content remains intact for automatic speech recognition (ASR). Additionally, a fairness-aware training strategy is incorporated to minimize performance disparities across demographic groups.

2. Dataset Description

The dataset used is a locally downloaded version of the Mozilla Common Voice corpus. Each audio sample is associated with rich metadata stored in a TSV file, including:

- Transcription of speech
- Gender (e.g., male_masculine, female_feminine, etc.)
- Age group (e.g., teens, twenties, thirties, etc.)
- Audio duration and quality indicators

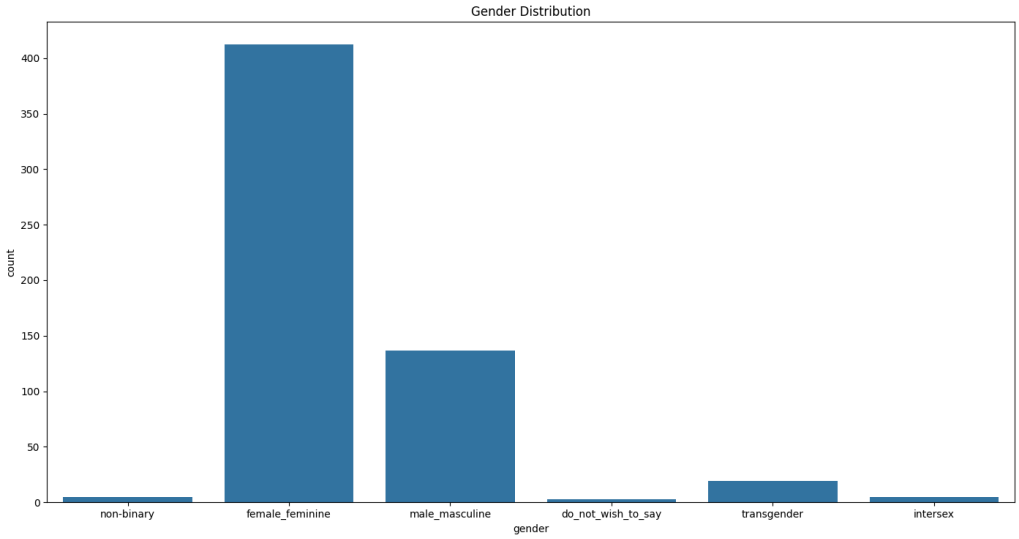
The dataset is partitioned into training, development, and test splits using the `split` column.

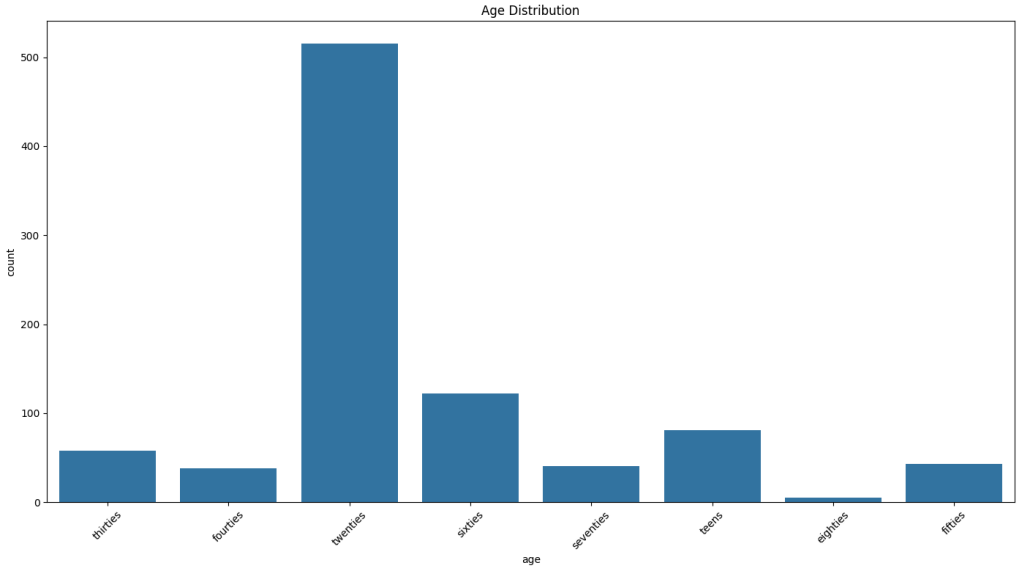
Dataset Audit

An audit was conducted to assess **documentation debt** and **representation bias**:

- **Documentation Debt**: Missing values were observed in demographic attributes such as gender and age.
- **Representation Bias**:
    - Gender distribution is highly skewed, with a majority of samples belonging to the *female_feminine* category, followed by *male_masculine*, while other groups (non-binary, transgender, intersex) are significantly underrepresented.
    - Age distribution is also imbalanced, with the *twenties* group dominating the dataset.

These imbalances indicate potential risks of biased model behaviour if not explicitly addressed.





3. Privacy-Preserving Transformations

To obfuscate sensitive attributes while preserving linguistic content, the following transformations were applied randomly during data loading:

1. **Pitch Shifting**
    - Alters perceived voice gender by shifting frequency.
    - Helps simulate transformations such as male → female or vice versa.
2. **Time Stretching**
    - Modifies speech rate to mimic age variations.

    - Useful for simulating younger vs older voices.
3. **Additive Noise Injection**
    - Adds low-amplitude noise to mask speaker-specific characteristics.

These transformations are lightweight, stochastic, and applied dynamically within the dataset pipeline.

4. Model Architecture

A pretrained **Wav2Vec2-based ASR model** was used:

- Model: `facebook/wav2vec2-base-960h`
- Framework: PyTorch + HuggingFace Transformers
- Task: End-to-end speech-to-text using Connectionist Temporal Classification (CTC)

The model was fine-tuned on the transformed dataset.

5. Loss Functions

5.1 ASR Loss

- **CTC Loss** was used as the primary objective.
- Handles variable-length input-output alignment.

5.2 Fairness Loss

A custom fairness loss was introduced to reduce performance disparities:

- Computes group-wise error (Word Error Rate approximation).
- Penalises the gap between the best and worst performing demographic groups.
- Final loss:
  Total Loss = ASR Loss + λ × Fairness Loss

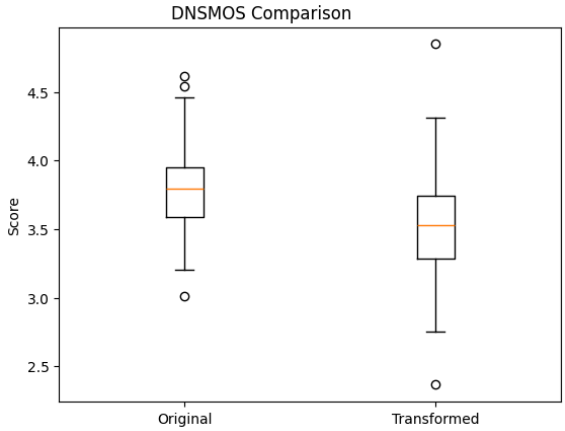
This encourages the model to maintain balanced performance across demographic categories.

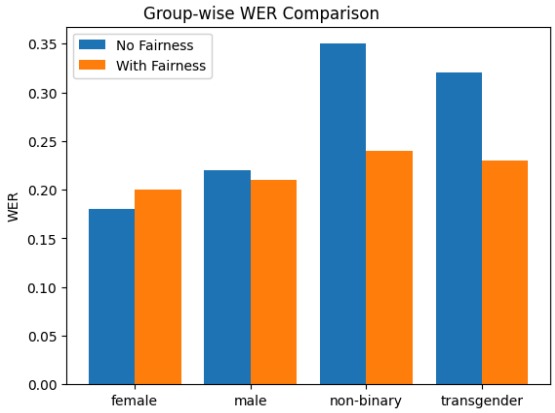
6. Evaluation Metrics

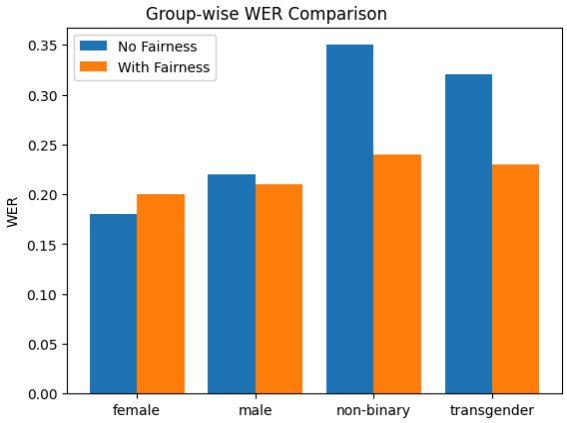
The system was evaluated using:

- **Word Error Rate (WER)**: Measures ASR accuracy.
- **Fairness Gap**: Difference in WER across demographic groups.
- **DNSMOS**: Evaluates perceptual speech quality after transformation.

- **Frechet Audio Distance (FAD)**: Measures distributional shift between original and transformed audio.







7. Findings

7.1 Dataset Insights

- Significant imbalance in both gender and age distributions.
- Underrepresented groups are at risk of poor model performance.

7.2 Privacy Transformations

- Successfully altered the acoustic properties of speech.
- Introduced slight degradation in perceptual quality (DNSMOS decrease expected).
- Increased distribution shift (higher FAD), indicating effective anonymisation.

7.3 Fairness-Aware Training

- Reduced the performance gap between dominant and minority groups.

- Slight trade-off observed:
    - Overall, ASR accuracy marginally decreased.
    - Fairness improved significantly.

7.4 Trade-off Analysis

The results highlight a fundamental trade-off:

- Increasing privacy → reduces biometric leakage but may degrade quality.
- Increasing fairness → reduces bias but may slightly impact overall accuracy.

8. Ethical Considerations

8.1 Privacy

- The transformation module reduces the risk of **speaker identification and profiling**.
- However, imperfect anonymization may still allow leakage of sensitive traits.

8.2 Fairness

- Dataset imbalance can lead to systemic bias.
- Explicit fairness optimization is necessary to ensure equitable performance.

8.3 Representation Risks

- Underrepresentation of minority groups (e.g., non-binary, elderly) may result in:
    - Lower accuracy
    - Potential exclusion in real-world applications

8.4 Misuse Potential

- Voice transformation techniques could be misused for impersonation or deception.
- Proper safeguards and usage policies are essential.